The empirical recurrence for OEIS A251296: recovering a conjecture that is not written down

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Abstract

OEIS A251296 records "Empirical recurrence of order 44", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1024$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 44, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 44 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A251296 is "Number of (n+1)X(4+1) 0..3 arrays with every 2X2 subblock summing to 3 6 or 9.". It has offset 1 and begins

$$12556, 164480, 2302844, 34973116, 579830068, 10472034724, 203797267488, 4207802278096, \ \ldots$$

The entry states:

> Empirical recurrence of order 44 (see link above)

As of the "Last modified" line on the live entry (Jun 02 2025 10:20:46, revision 6) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1024$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 1024$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2048$ exact terms of the model returns order 44 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{44} c_i\, a(n-i),$$

c_1	180	c_{12}	181825477805808	c_{23}	−796334522653748968267	c_{34}	16550686614649
c_2	−14781	c_{13}	5484452193315518	c_{24}	717162169212450235554	c_{35}	−3032389334010
c_3	730630	c_{14}	−21986222587047779	c_{25}	2897236666963920612541	c_{36}	−4788578438364
c_4	−24111595	c_{15}	−72254617451472756	c_{26}	256247852278506573548	c_{37}	8285536638143
c_5	555086319	c_{16}	535474117113796655	c_{27}	−5268639324464830040421	c_{38}	8445869027358
c_6	−8985783552	c_{17}	652832651408939750	c_{28}	−2981015195574035215362	c_{39}	−185088377555
c_7	98949942605	c_{18}	−7673158167846990497	c_{29}	5157235191948489437368	c_{40}	−774824098883
c_8	−647796156160	c_{19}	−7365962704206630516	c_{30}	4627562754563976489169	c_{41}	2396240766872
c_9	882754215177	c_{20}	71006846077096892525	c_{31}	−2922582551326018029662	c_{42}	177978424445
c_{10}	24908280360482	c_{21}	100259716410555978251	c_{32}	−3588426338200083292608	c_{43}	−11592827456
c_{11}	−203494191374813	c_{22}	−363807490822335475038	c_{33}	1060916892387494398000	c_{44}	99414953164

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 44, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $44 + 4$ terms removed returns order 44 as well, so $\deg q_{\min} = 44 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $44 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 44 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A251296.

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